School of Computing & IT
Web Programming (CIT2011)
Week 1 – Exploring the Web with Developer Tools

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1. Lab Overview

Modern web developers need to understand what happens inside the browser when a webpage loads and when a user interacts with it. In this lab, students use Chrome, or Firefox Developer Tools to inspect HTML, and CSS.

Register to https://onecompiler.com/. You will use this online compiler to share your ongoing labs with your tutor for review.

2. Learning Outcomes

1. Openandnavigate browser Developer Tools.
2. Identify HTML elements using the Elements/Inspector panel.
3. Modify HTML and CSS temporarily using Developer Tools.
4. Use responsive/device emulation to test different screen sizes.
5. Use Developer Tools as part of a modern web-development workflow.

3. Software Requirements

Studentsneedthefollowing:

- Google Chrome / Mozilla Firefox
- Internet connection
- A text editor such as VS Code

4. Developer Tools Overview

Chrome	Firefox	Purpose
Elements	Inspector	Examine HTML and CSS
Console	Console	Execute JavaScript and view errors
Sources	Debugger	Examine JavaScript/source files
Network	Network	Examine HTTP requests
Application	Storage	Examine cookies/storage
Lighthouse	Lighthouse	Performance/accessibility analysis
Device Toolbar	Responsive Design Mode	Test different screen sizes

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PART A — Opening Developer Tools
Activity 1: Open Developer Tools

a. Open https://www.w3schools.com/html/default.asp or another publicly accessible website of your choice).
b. Open Developer Tools using right-click → Inspect, F12, Ctrl+Shift+I on Windows, or Cmd+Option+I on macOS.

Question 1. What is the name of the panel used to inspect the webpage's HTML? Elements

Question 2. List five panels available in Developer Tools.

1. Elements 2. Source 3. Console

4. Network 5. Performance

PART B — Inspecting HTML
Activity 2: Examine the HTML Structure
Selectthe Elements/Inspectorpanel.Locate<html>, <head>, <body>, headings, paragraphs, links, images and navigation elements.
Question 3. Identify one heading element. Record the HTML tag and the text inside it.
HTML tag: <h1> </h1>

Text: Learn To Code
Question 4. Find one hyperlink. Record the HTML element and its href value.
HTML element: C++

href: href=’/cpp/default.asp’

Activity 3: Modify HTML
Findavisibleheading.Right-click the element and select Edit as HTML (or double-click it). Temporarily change it

to:

<h1>CIT2011 Developer Tools Lab</h1>
Question 5. What happened to the webpage?
The website heading changed to the text I edited it to in Developer tool
Question 6. Was the original website permanently changed? Explain why.
The original website didn’t change because once I refreshed it went back to normal.
You're usually only changing the copy of the webpage that exists inside your browser and I don’t believe the main one is something everyone should be able to access.

PART C — Inspecting CSS

Activity 4: Examine CSS

Selectan HTMLelementcontaining visible text. Examine the Styles panel and identify CSS properties controlling font size, font family, margin, padding, text colour and background colour.

Property	Value
font-size	21px
font-family	Source Sans Pro", "Segoe UI", sans-serif
margin	0 0 20px;
padding 1.	10px 20px;
color 2.	white#fff
background-color	#282A35 !important;

Activity 5: Modify CSS

Addthe followingdeclaration to a heading in the Styles panel:

font-size: 50px;

Then change it to:

font-size: 20px;

Question 1. What happened when the font-size value changed?

It got a bit smaller since it was original at 62

Question 2. Add text-transform: uppercase; What happened?

The entire text became only capital letters

Question 3. Change margin and padding. Explain the difference between margin and padding.

Padding creates space between the content and the border and Margin creates space between the element and other elements

PART D — Understanding the Box Model

Activity 6: Inspect the Box Model

Select an element and locate the Box Model diagram.

Question 4. Identify the four major components of the CSS box model.

1. Border 2. Content 3. Margin 4. Padding

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PART I — Responsive Web Design

Activity 15: Device Emulation

Open Device Toolbar / Responsive Design Mode and test mobile, tablet, laptop, and desktop sizes such as 375×667, 768×1024, and 1366×768.

Question 5. What happened to the webpage when the screen width became smaller?

The webpage on a whole became more compact and certain options went away

Question 6. Did the navigation change? Describe what happened.

Navigation indeed changed, I was able to notice that the titles at the top were now available in a drop down/ladder menu

Question 7. Did images resize? Explain your observation.

Yes, images resized in order to accomodate for visitors that or on various different devices which has different resolutions

Activity 16: Test Your Own Responsive Design

Create a file named responsive.html using the following code:

```
<!DOCTYPE html>
<html>
<head>
  <title>Responsive Test</title>
  <style>
    body { font-family: Arial; margin: 20px; }
    .container { display: flex; gap: 20px; }
    .box { padding: 30px; background-color: lightblue; flex: 1; }
    @media (max-width: 600px) {
      .container { flex-direction: column; }
    }
  </style>
</head>
<body>
  <h1>Responsive Web Design</h1>
  <div class="container">
    <div  class="box">Box  1</div>
    <div  class="box">Box  2</div>
    <div class="box">Box 3</div>
  </div>
</body>
</html>
```

Use Developer Tools to test 1200px, 900px, 700px, 600px, 500px and 375px.

Question 8. At approximately what width does the layout change from rows to columns?

600px

Question 9. Which CSS feature caused the layout to change?

flex-direction: column;

Question 10. Complete the table above.

FINAL CHALLENGE — Activity 20: Become the Web Detective

A student reports that a webpage is displaying incorrectly on their mobile phone, although it works correctly on a laptop. Use Developer Tools to investigate responsive layout, CSS rules, media queries, images, JavaScript errors, network requests, and HTTP status codes.

Question 11. Identify three possible reasons why a webpage may work correctly on desktop but incorrectly on mobile.

1. Incorrect or missing CSS media queries for mobile screen sizes.
2. Images with fixed dimensions that do not fit the smaller screen.
3. Text is too large

FINAL REFLECTION

Question 12. How does Developer Tools help a web developer understand a webpage?

Developer Tools help web developers understand how a webpage is constructed while giving them options to troubleshoot HTML, CSS, JavaScript etc...

Question 13. Why is it useful to inspect HTML and CSS rather than simply looking at the webpage?

It allows us to identify errors, understand why elements appear a certain way, and make changes or fix problems more easily.

Question 14. Why is the Network panel important to a web developer?

It allows us to see the files and resources a webpage loads. It also helps identify failed requests and loading problems.

Question 15. Why is responsive testing important?

Responsive testing is important because it ensures that a webpage displays and functions correctly

Question 16. How can Developer Tools help identify bugs?

Developer Tools will help you to identify bugs by allowing you to inspect HTML and CSS, view JavaScript errors, check network requests, and test changes in real time.

Question 17. Explain how HTML, CSS and JavaScript work together to create an interactive webpage.

HTML provides the structure, CSS controls the appearance, and JavaScript adds interactivity to create a functional and engaging webpage.

Question 18 – Build a webpage

Use thetemplateprovided byyour tutor to build a webpage using HTML5 and basic CSS3. Webpage must include, basic HTML5 tags such as, image, paragraph, navigation, heading, sections, footer, links, etc.

Practical Deliverables

- Completed lab sheet (Questions 1–18).
- At least six screenshots: HTML inspection, CSS inspection
- responsive.html